

TRANSCRIPTIONS

Data Collection & Analysis Assignment - Ben Dignan P6

See individual transcriptions in subtabs on the left-hand side

Dr. Ford

Interview with Licensed Physical Therapist

The participant, Dr. Ford, PT, is a licensed physical therapist practicing in a primarily hands-on, manual therapy setting. The interview was conducted via text message since she was at a convention in South Carolina.

BD: From your clinical experience or independent research, have you found AI tools like movement analysis software or predictive recovery models to be helpful in patient assessment? Where do they still fall short?

TF: I have not used these models in my practice.

BD: In your opinion, what underappreciated aspects of therapy require uniquely human skills and to what scale do you foresee AI complementing those?

TF: I practice with an extremely hands-on approach as a manual therapist. AI will not be able to replicate those skills.

BD: What ethical concerns do you have as AI becomes integrated in the field, especially in terms of accuracy/patient safety and professional responsibility?

TF: I feel like the human aspect of PT is very important and, in terms of treatment, AI will not replicate this. With respect to documentation, I do feel AI could be very helpful for taking notes while I perform my interviews or through dictation. Proof reading will be necessary to ensure accuracy.

BD: Based on your professional experience, what would you consider before deciding whether an AI tool is credible or worth incorporating into practice?

TF: Cost, accuracy, ease of use.

BD: How do you imagine AI reshaping PT or PT tech roles over the next decade? Will it expand the therapist's capabilities, replace some tasks, redefine, etc.?

TF: It could only help with time management if accurate, allowing PTs more time with patients and less time documenting.

BD: Do you think AI could improve accessibility to physical therapy for underserved patients? Why or why not?

TF: Possibility from an exercise aspect.

BD: If you could design a dream AI tool to accompany you, what would it do and what should it not do?

TF: It would write my notes for me

Ms. DeSola

Interview with Grad Student

The participant, Ms. DeSola, is pursuing a DPT at Northwestern University. The interview was conducted back and forth on messages.

BD: In your opinion, what underappreciated aspects of therapy require uniquely human skills and to what scale do you foresee AI complementing those?

DD: I think that underappreciated aspects of therapy that can only be understood by humans include emotional and social skills. As PTs, we have to be prepared for difficult situations or conversations (ex. Patient recently diagnosed with cancer, can't afford therapy because they lost their job, etc.) that require caring and considerate communication as well as empathy. AI could be used by therapists to practice these skills. For example, using ChatGPT you can ask it to pretend to be a patient who is experiencing a tough time. This will allow the therapist to practice carrying out that conversation and increases the likelihood of caring and professional responses in real life situations.

BD: Have you learned about any AI or digital tools that change the way patients complete their rehabilitation process? If so, which ones and how effective were they?

DD: No I have not.

BD: Have your professors employed the use of AI in the classroom or in demonstrations?

DD: Yes, our professors give us questions to work on as a group during lecture. After we have discussed our answers, we are told to "ask AI" to see if it gives us additional ideas or confirms our answers.

BD: What ethical concerns do you have as AI becomes integrated in the field, especially in terms of accuracy/patient safety and professional responsibility?

DD: I know some clinics are already using AI to save time on patient documentation. This creates some concern because if not double checked, these notes could include inaccurate information or miss key details regarding a patient's case or condition. It is crucial to have accurate documentation to be able to safely treat each patient with precautions if necessary.

BD: What would you consider before deciding whether an AI tool is credible or worth incorporating into practice?

DD: I'd consider researching/reading about its credibility. I would also consider the credibility of people or corporations that use the same AI tools.

BD: How do you imagine AI reshaping PT or PT tech roles over the next decade? Will it expand the therapist's capabilities, replace some tasks, redefine, etc.?

DD: I can very easily see documentation become more efficient and less time consuming with AI.

BD: Do you think AI could improve accessibility to physical therapy for underserved patients? Why or why not?

DD: It could. Maybe by using AI to identify areas that are lacking PT clinics. This information can be used to pinpoint locations in which opening a PT clinical would be maximally beneficial.

BD: If you could design a dream AI tool to accompany you, what would it do and what should it not do?

DD: I'd create an AI tool that has access to the information from my curriculum (ex. Textbooks, quizlets) that would consolidate important considerations when I am looking at a specific patient case.

Dr. Khan

Dr. Fayaz Khan is a professor at King Abdulaziz University in Saudi Arabia. He has published many medical papers about AI's role in rehabilitation.

What are the most promising areas of research involving AI in physical therapy or rehabilitation science today?

The most promising areas involving AI in Physical therapy would be in prediction of outcome for a long time, particularly in neurological conditions like stroke. AI can also be beneficial in tele rehabilitation and monitoring the outcome using AI integrated systems.

Do you utilize AI in your profession and is it part of what you teach? If so, what aspects do you use?

We are working on the basics in integrating AI in research, we just tried to know the knowledge and attitudes of physical therapists on using AI. Used a predictive model for detecting lumbar disc prolapse

How has AI changed the way researchers analyze human movement and rehabilitation outcomes?

Yes, AI can be utilised in analysis of movement which can be incorporated with motion analysis labs. The data can be validated and interpreted using the AI.

Many AI systems rely on large datasets of patient movement and recovery outcomes. What challenges exist when collecting or training AI models using rehabilitation data?

Exactly, this is now a big challenge in the field of rehabilitation. Unlike other medical systems the data sets are rarely maintained in detail in rehabilitation departments. Infact data collection for rehabilitation is not that easy compared to other medical specialties as it involves many condition based outcome measures which is extensive.

How reliable are current AI-based movement analysis tools compared to traditional assessment by trained therapists?

The studies on reliability are not consistent yet, it ranges from 60-90% accuracy. Hence I need more studies to come to a conclusion.

What limitations currently prevent AI from replacing clinical reasoning performed by physical therapists?

That's a good question, as far as my experience none of the AI based tools are not competent in clinical reasoning as compared to physical therapists. It needs more accuracy, sophistication and comprehensiveness including multiple variables.

How has AI improved documentation systems such as SOAP notes or electronic medical records in physical therapy practice, and what is still left for the future?

As mentioned in the previous answer for question 4, in the field of physical therapy there are many outcomes which are intensive. Physical therapists are not dependent alone on the SOAP notes.

Some researchers have raised concerns about bias in medical AI systems. How might bias affect AI tools used in rehabilitation settings?

In rehabilitation AI tools need more validation and more data for testing.

What role do you think physical therapists should play in guiding the development and ethical use of AI technologies in healthcare?

First focus should be to provide more accurate data which are focussed for particular clinical conditions. It is not good to rely on fancy AI tools which give false hopes.

How do you personally envision AI to influence common clinical practices over the next decade?

AI would play a great role in the outcome prediction and planning the treatment, which in turn can save manpower and income.

Dr. Plotkin

Dr. Plotkin is a Doctor of Chiropractic, Certified Strength and Conditioning Specialist, and Certified Sports Nutritionist who has worked with athletes at the highest levels. He has served as the Team Chiropractor for the University of Maryland Terrapins since 2005, helping college athletes stay healthy and perform at their best. He also worked with the Baltimore Orioles gaining experience supporting professional athletes. In addition to his clinical work, Dr. Plotkin teaches in the Kinesiology Department at the University of Maryland, where he instructs courses on muscle function and exercise rehabilitation. He has experience in sports medicine, chiropractic care, strength training, and athletic performance.

Do you utilize AI in your profession and is it part of what you teach? If so, what aspects do you use?

How has AI changed the way researchers analyze human movement and rehabilitation outcomes?

How reliable are current AI-based movement analysis tools compared to traditional assessment by trained therapists?

What limitations currently prevent AI from replacing clinical reasoning performed by physical therapists?

How has AI improved documentation systems such as SOAP notes or electronic medical records in physical therapy practice, and what is still left for the future?

How might bias affect AI tools used in rehabilitation settings and how are physical therapists obligated in guiding the development and ethical use of AI technologies in healthcare? How can we ensure bias free and ethical results?

How do you personally envision AI to influence common clinical practices over the next decade?

AI is something that if we don't learn to use in our favor we will be left behind. That being said, our jobs are safe as far as hands-on work that only a human can do! We do use it to help with typing reports, sometimes letters of medical necessity etc. In the future I see it helping a ton with paperwork, billing follow up, treatment notes etc.

Mr. Mariano

Mr. Marciano Mariano has served as an advisor to the Obama administration and likes keeping up to date with the latest AI trends.

What are the most promising areas of research involving AI in physical therapy or rehabilitation science today?

AI works most efficaciously when domain data abounds, and its capture and transfer structured with standards. This is certainly the case with the PT and rehabilitation industry. The most promising areas are those with the greatest patient and stakeholder value such movement analysis, fall-risk prediction, stroke and orthopedic rehabilitation, wearable sensors, robotic rehab devices, remote monitoring, personalized exercise plans, and AI-assisted outcome prediction. AI is especially strong where there is a lot of motion data: gait, posture, range of motion, balance, recovery patterns, and exercise compliance.

How has AI changed the way researchers analyze human movement and rehabilitation outcomes?

AI has changed research by making it possible to analyze movement at a much finer level than the human eye alone can manage. Video, wearables, force plates, phone sensors, and motion-capture systems both collect and operate from data available to be analyzed for patterns that might predict pain, weakness, compensation, fatigue, or recovery progress.

Many AI systems rely on large datasets of patient movement and recovery outcomes. What challenges exist when collecting or training AI models using rehabilitation data?

Rehabilitation data is inherently difficult, as patients vary by age, injury, motivation, pain tolerance, disability, culture, home environment, and access to care. Data may be incomplete, inconsistent, or collected with different devices. Also, medical data is private and must be protected. A model trained on one population may not work well on another. Further privatization standards between AI systems is still immature driving understandable concerns that personal data handled correctly in one AI system may be compromised in the succeeding AI system.

What limitations currently prevent AI from replacing clinical reasoning performed by physical therapists?

Quite simple, context and human-prioritized insights. Physical therapy is more than just pattern recognition. A therapist reads the whole person: pain, fear, effort, family support, confidence, frustration, goals, and subtle changes from visit to visit. AI may suggest, measure, or predict, but

it does not truly understand the patient's lived experience. Clinical reasoning includes judgment, ethics, communication, and adaptation in real time.

How has AI improved documentation systems such as SOAP notes or electronic medical records in physical therapy practice, and what is still left for the future?

AI is already helping with documentation by drafting SOAP notes, summarizing visits, organizing EMR data, and reducing repetitive typing. The future will be better integration: AI that listens, structures notes, flags missing information, and connects treatment plans to outcomes. But clinicians must still verify the record. A wrong note is still a clinical and legal problem

Some researchers have raised concerns about bias in medical AI systems. How might bias affect AI tools used in rehabilitation settings?

Bias could appear if the training data underrepresents older adults, disabled patients, rural patients, racial groups, women, non-English speakers, or people with unusual movement patterns. Then the AI might make poorer recommendations for exactly the patients most in need of careful care.

How are physical therapists obligated in guiding the development and ethical use of AI technologies in healthcare? How can we ensure bias-free and ethical results?

Physical therapists must help shape these systems, not simply receive them from technology companies. They should ask: Was the data representative? Is the model explainable? Does it protect privacy? Has it been tested on real patients? Does it improve care, or just improve billing and paperwork? Bias-free is probably impossible, but bias-aware, bias-tested, and ethically governed is necessary.

How do you personally envision AI to influence common clinical practices over the next decade?

Over the next decade, I expect AI to become a quiet and integrated tool and clinical partner. It will help measure movement, document visits, monitor home exercise, predict recovery risk, and personalize rehab plans. But the best therapists will still be the ones who combine technology with human discernment. The machine may see the pattern, but the therapist must still see the person.

Additional Comment: Wearables and AI make sense when they (probably the critical success factors for all technology use):

save time

improve outcomes

or let therapists manage more patients

But clinics don't adopt tech just because it's cool. It has to fit into workflow, integrate with records, and get reimbursed by insurance. That's where things are still catching up.

Right now:

Tech is good enough

Prices are coming down

But systems (insurance, EMRs, workflows) are the bottleneck

Over the next 5–10 years, I expect these tools to become standard (and because they become more standardized)—not because they got dramatically smarter, but because they got cheaper, easier, and better integrated.

SUMMARIES

Interview 1: Dr. Ford (Practicing PT)

Dr. Ford reported that she does not currently use AI in her clinical practice. She strongly emphasized the importance of hands-on manual therapy, stating that AI cannot replicate these physical skills.

She acknowledged that AI could be helpful for documentation, such as taking notes or dictation, but emphasized the need for proofreading to ensure accuracy.

Overall, she believes AI may help with time management but will not significantly change patient treatment, reinforcing the importance of human interaction in therapy.

Interview 2: Ms. DeSola (DPT Student)

Ms. DeSola highlighted the importance of human skills in therapy, especially empathy, communication, and handling difficult patient situations. She noted that AI could actually help therapists practice these interpersonal skills through simulation tools.

She also pointed out ethical concerns, particularly with AI-generated documentation, which may contain inaccuracies if not reviewed. She believes AI will significantly improve efficiency, especially with note-taking.

Additionally, she suggested AI could improve accessibility by identifying underserved areas. Her perspective reflects optimism about AI as a support tool rather than a replacement.

Interview 3: Dr. Khan (Professor/Medical Researcher)

Dr. Khan focused on AI's research potential, especially in outcome prediction for neurological conditions like stroke and in tele-rehabilitation. He noted that AI can assist in analyzing movement data and integrating with motion labs.

However, he emphasized that rehabilitation faces major data collection challenges due to inconsistent and complex patient data. He also stated that AI tools are not yet reliable enough (60–90% accuracy) to replace therapists.

He stressed that AI lacks the sophistication needed for clinical reasoning and requires further validation. He also warned against over-reliance on “fancy AI tools” without strong supporting data. Overall, he believes AI will improve planning and efficiency but still requires human oversight.

Interview 4: Dr. Plotkin (Sports Chiropractor/Professor)

Dr. Plotkin expressed a practical view of AI, stating that professionals who fail to adapt may fall behind. He emphasized that AI is already useful for administrative tasks such as reports, billing, and documentation.

However, he was clear that hands-on care cannot be replaced, especially in fields requiring manual therapy and physical interaction. He believes AI's primary future role will be improving efficiency rather than replacing clinicians.

Interview 5: Mr. Mariano (Former White House Advisor/AI Researcher)

Mr. Mariano emphasized that AI is most effective in physical therapy when large amounts of structured data are available. He identified key applications such as movement analysis, fall-risk prediction, stroke rehabilitation, wearable technology, and personalized exercise planning. He explained that AI allows researchers to analyze movement in far greater detail using tools like motion capture, sensors, and video data.

However, he noted major challenges, including inconsistent data, privacy concerns, and lack of standardization across systems. He stressed that AI cannot replace clinical reasoning because therapists consider emotional, psychological, and contextual factors that AI cannot interpret.

He also highlighted AI's growing role in documentation (SOAP notes, EMRs), but warned that clinicians must verify all outputs. Looking forward, he believes AI will become an integrated tool that improves efficiency, but therapists will remain essential because "the machine may see the pattern, but the therapist must still see the person."

PATTERNS

Key Patterns and Themes

Theme	Key Evidence / Quotes	Pattern Observed
AI improves efficiency	“Reduce note-writing time by 50–70%” (Mariano) “helping with paperwork, billing, notes” (Plotkin)	Strong agreement: AI is best for administrative tasks
AI cannot replace therapists	“AI does not understand the patient’s lived experience” (Mariano) “AI will not replicate hands-on skills” (Ford)	Universal agreement across all interviews
Importance of human skills	“Empathy and communication are essential” (DeSola)	Human interaction is irreplaceable
Data challenges	“Data is inconsistent and difficult to collect” (Mariano, Khan)	Major limitation in AI development
Accuracy concerns	“60–90% accuracy” (Khan) “must verify notes” (Ford)	AI is not fully reliable yet
Future role of AI	“Integrated tool...clinical partner” (Mariano) “help with efficiency” (Plotkin, DeSola)	AI seen as assistive, not replacement
Ethical concerns	Bias, privacy, incorrect documentation (multiple respondents)	Consistent concern across interviews

DATA ANALYSIS

Written Data Analysis

A clear pattern emerges across the five interviews: AI is seen as a tool to support, not replace, physical therapists. All respondents, regardless of their background, emphasized that human skills like clinical reasoning, empathy and manual treatment cannot be replaced by AI. This consistency gives strong support to the hypothesis that AI will augment, but not replace, physical therapy practice. A big theme running through the responses is that the biggest strength of AI is efficiency and administrative support. Several participants mentioned it was helpful in documentation (e.g., SOAP notes, electronic medical records), billing, and recordkeeping. This means AI is already influencing the profession by reducing time spent on repetitive tasks, so therapists can spend more time on patient care.

That being said, several limitations to AI were repeatedly brought to light. One of the greatest concerns is data quality and availability, particularly in rehabilitation settings where patient data is inconsistent. There are also concerns about accuracy, with estimates of reliability being 60-90%, according to Dr. Khan. Additionally, with AI arises questions about bias and privacy. Overall, AI is not reliable enough to be used for clinical decisions on its own. Such concerns underline the need for continuous human oversight in all applications of AI in physical therapy. Perceived differences in perspective based on professional background were also seen. For example, researchers like Dr. Khan and Mr. Mariano tended to emphasize the technical capabilities and long-term potential of AI, while practicing clinicians like Dr. Ford emphasized current limitations and lack of integration. The contrast can best be defined that AI has promise in research settings, but its practical implementation is underdeveloped.

From this data, it can be concluded that AI will likely become an assistive tool that enhances efficiency, improves data analysis, and expands access to care, but it will not replace therapists due to the importance of human judgment and interaction, directly supporting my original hypothesis. Nonetheless, there are limitations of the research conducted. The sample size is relatively small with five interviews. While I was successful in reaching professionals with unique backgrounds, more voices would solidify these patterns more effectively. Moreover, two of the respondents have limited experience with AI, which provides a reality check, but may influence the depth of their insights. This research also raises more questions about AI's future in physical therapy, particularly regarding how data collection can be improved, how bias can be reduced, and how fast AI tools will be integrated into regular clinical practice. The findings indicate that AI is an important but supportive component of the evolution of physical therapy.